

The block-rank theorem for Fermat CRT blocks, and the red link that makes it unconditional

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Abstract

Let (n, m) be an even-dimensional Fermat cell and $p \mid m$ a prime. Over $\mathbb{F}_p$ the cyclotomic quotient splits, by the Chinese Remainder Theorem, into blocks; the rank a block contributes at coupling depth ℓ obeys the recursion $\text{rec}(d, 0) = 1$, $\text{rec}(d, 1) = d$, $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1)$ for $\ell \geq 2$, where $\text{DS}_n(m)$ is the Degtyarev–Shimada complex rank polynomial. We prove this identity holds *for every* d , by a chain of module-dimension identities read from the Degtyarev–Shimada source: the measured block is a Degtyarev–Shimada core module (Cor. 1.7, free tail), the core dimension is a closed lattice walk (Remark 4.4), the walk is the Degtyarev–Shimada value, uniform in d by Lipshitz finiteness. Version 1.0 stated this conditional on one identification (the *dictionary*): that the measured CRT all- A block is the Degtyarev–Shimada core. The present version discharges that condition. The dictionary has two halves: a geometric identification ($e_A \leftrightarrow$ coordinate-zeroing), confirmed textually against the source, and a *faithfulness* claim — that the $\mathbb{F}_p$ refinement of the integral block injects no spurious torsion. We prove faithfulness by a pen-and-paper, characteristic-independent argument: the block closure is a free $\mathbb{Z}$ -module (it is the ring $\mathbb{Z}[t]/\varphi$ realised inside the pair, characteristic polynomial φ), and a free module reduces faithfully modulo every prime. The block-rank law is therefore **unconditional**. The integral-torsion question (Degtyarev–Shimada Conjecture 1.2) sits outside this chain and remains open: this theorem fixes the per-block *value*, not whether torsion exists.

1 What changed from Version 1.0

Version 1.0 proved the block-rank law $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1)$ *conditional* on one identification, the *dictionary* of §2: that the campaign’s measured CRT all- A block is the Degtyarev–Shimada core. That dictionary has two halves.

- **(geometric)** $e_A \leftrightarrow$ the coordinate-zeroing cut $X(2s)$ of [1, §4.6]. This half was *verified textually* against the source — never the open part.
- **(faithfulness)** that the block measured over $\mathbb{F}_p$ equals the core’s reduction over $\mathbb{Z}$ *without spurious torsion injected when $\mathbb{Z} \rightarrow \mathbb{F}_p$* . This was the genuinely open link, carried until now by measurement rather than proof.

The present version closes the faithfulness half by a pen-and-paper, characteristic-independent proof (§6). The block closure is shown to be a free $\mathbb{Z}$ -module, and a free module reduces faithfully modulo every prime. With the geometric half textual and the faithfulness half a theorem, the dictionary

is discharged and **the block-rank law is unconditional**. The swan (integral torsion) still sits outside the chain and remains open.

2 Setting and statement

Fix an even-dimensional Fermat cell (n, m) , $n = 2d$, and a prime $p \mid m$. The integral Hodge question for the standard d -spaces reduces, by Degtyarev–Shimada [1], to whether $\dim_{\mathbb{C}}$ of the cycle lattice equals $\dim_{\mathbb{F}_p}$ for every $p \mid m$; the complex rank is the closed-form polynomial $\text{DS}_n(m)$ of [1, Remark 4.4]. Over $\mathbb{F}_p$ the per-variable cyclotomic ring splits, by the Chinese Remainder Theorem, into a $(t-1)$ -primary factor A and a coprime cofactor B ; tensoring over the $n+2$ variables decomposes the cohomology module into CRT blocks. The rank a block contributes at *coupling depth* ℓ (the number of A -pairs in the block) was measured to obey

$$\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1) \quad (\ell \geq 2), \quad \text{rec}(d, 1) = d, \quad \text{rec}(d, 0) = 1,$$

the General Product Law. The mechanism by which the ladders *mesh* as ℓ grows was a measured pattern. We prove it.

Theorem 1 (Block-rank law, unconditional). *For every $d \geq 0$ and $\ell \geq 1$,*

$$\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1).$$

Equivalently, the rank the block contributes at depth ℓ is the constant term of a closed lattice walk of 2ℓ steps on $\mathbb{Z}^{\lfloor d/2 \rfloor}$, with rest permitted iff d is odd.

The published Degtyarev–Shimada closed forms, with $\delta_m = (m-1) \bmod 2$, are $\text{DS}_2(m) = 3m^2 - 9m + 6 + \delta_m$, $\text{DS}_4(m) = 15m^3 - 90m^2 + 175m - 100 + (15m - 39)\delta_m$, and $\text{DS}_6(m) = 105m^4 - 1050m^3 + 3955m^2 - 6335m + 3325 + (210m^2 - 1302m + 2010)\delta_m$; these are the only rows for which [1] prints a polynomial. The theorem covers all $n = 2(\ell-1)$ by the closed-walk reading of §3, which is the *definition* of DS_n for $n > 6$.

3 The dictionary — stated, and discharged in §6

The theorem identifies the campaign’s measured object with a Degtyarev–Shimada module. We record the identification in full, with its two halves separated, because the separation is exactly what a referee needs to verify the discharge.

The dictionary. The campaign’s CRT all- A block at depth ℓ is the Degtyarev–Shimada core module $\overline{\mathcal{C}}_{J(2s)}(2s)$, $s = \ell - 1$. Concretely: the A -idempotent e_A (projecting onto the $(t-1)$ -primary part over $\mathbb{F}_p$) corresponds to the coordinate-zeroing $z_v = 0$ of the partial Fermat $X(2s) := W_s \cap \{z_{2s+2} = \cdots = z_{n+1} = 0\}$ [1, §4.6]; the live cofactor e_B ($a_i \neq 1$) corresponds to the surviving coordinates of Γ_K [1, Def. 1.3, Thm. 1.4].

This dictionary has two halves, and only one was open:

- **(geometric, textual)** the identification $e_A \leftrightarrow X(2s)$ itself, confirmed *verbatim* against the Degtyarev–Shimada text (Definition 1.3’s “ $a_i \neq 1$ ” is e_B ; “ $a_i = 1$ ” is e_A ; the partial-Fermat cut $X(2s)$ of §4.6 is the coordinate-zeroing). A textual reading, not an inference.
- **(faithfulness, open)** that the block measured over $\mathbb{F}_p$ equals the core’s reduction over $\mathbb{Z}$ without spurious torsion injected in the passage $\mathbb{Z} \rightarrow \mathbb{F}_p$. The Degtyarev–Shimada tensor is over $\mathbb{Z}$; the CRT split is over $\mathbb{F}_p$ and carries the nilpotent $(t-1)$ -primary tower (multiplicity

$p^v - 1$ over $\mathbb{F}_p$, multiplicity 1 over $\mathbb{C}$). Faithfulness is the claim that this tower injects no extra survivors.

§6 proves the faithfulness half. With the geometric half textual and the faithfulness half a theorem, the dictionary is discharged and the block-rank law of §2 is unconditional.

4 Link 1 — the value is a closed walk

Proposition 1. $DS_n(m)$ is the number of closed walks of $n + 2$ steps on $\mathbb{Z}^{\lfloor (m-1)/2 \rfloor}$ (each step ± 1 along an axis, returning to the origin), with a rest step permitted iff m is even. Equivalently it is the constant term of $(1 + S)^{n+2}$ (even m) or S^{n+2} (odd m), where $S = x_1 + \dots + x_h + x_h^{-1} + \dots + x_1^{-1}$ is the symmetric Laurent sum.

This is read directly from [1, Remark 4.4], gate-verified byte-exact on 24 in-sample points (DS_2, DS_4, DS_6 for $m = 3..8$), 12 out-of-sample points ($m = 9..12$), the row DS_8 , and the decisive independent check $DS_6(4) = 1107$, measured on the author's hardware and reproduced by the formula without having been fitted to it. The parity correction δ_m is exactly the walker's rest permission. With $s = \ell - 1$, $m = d + 1$, the depth- ℓ value $DS_{2(\ell-1)}(d + 1)$ is a closed walk of 2ℓ steps on $\mathbb{Z}^{\lfloor d/2 \rfloor}$, rest iff d is odd.

5 Link 2 — the measured block is the core, and the core is the walk

The equality is carried by a tensor identity, *not* by a shared recursion (two sequences being holonomic does not make them equal).

Lemma 1 (block rank = core dimension). *Under the dictionary of §3, the rank of the measured block at depth ℓ equals $\dim_{\mathbb{C}} \overline{C}_{J(2s)}(2s)$, $s = \ell - 1$.*

Proof. By [1, Cor. 1.7], the module decomposes as a tensor

$$\overline{C}_{J_s}(2d) = \overline{C}_{J(2s)}(2s) \otimes_{\mathbb{Z}} \overline{S}(s, d), \quad \overline{S}(s, d) = \bigotimes_k \mathbb{Z}[t_k]/(\varphi(t_k)),$$

each tail factor $\mathbb{Z}[t_k]/(\varphi(t_k))$ free of rank $m - 1$, hence $\overline{S}$ free. The exact sequence $0 \rightarrow (\lambda)/(\lambda\rho_J) \rightarrow R/(\lambda\rho_J) \rightarrow R/(\lambda) \rightarrow 0$ with $R/(\lambda)$ free [1, Lemma 4.5, Lemma 4.1] makes the split clean. Because $\overline{S}$ is free of rank $(m - 1)^{d-s}$, over $\mathbb{C}$ the tensor dimension factorises as $\dim_{\mathbb{C}} \overline{C}_{J_s}(2d) = \dim_{\mathbb{C}}(\text{core}) \cdot (m - 1)^{d-s}$, the tail contributing only its constant free multiplicity. The rank the block contributes at depth ℓ is therefore $\dim_{\mathbb{C}}$ of the core exactly. $\square$

Lemma 2 (core dimension = walk). $\dim_{\mathbb{C}} \overline{C}_{J(2s)}(2s)$ is the closed walk of $2s + 2$ steps of Proposition 1.

Proof. The core is the full-partition module one dimension down: by [1, §4.6], the restriction of J_s to $\{0, \dots, 2s + 1\}$ is exactly the full set $J(2s)$ of partitions of $2s + 1$. Remark 4.4 applies to the core verbatim with $n \rightarrow 2s$, so $\dim_{\mathbb{C}}(\text{core}) = DS_{2s}(m)$, the constant term of $(1 + S)^{2s+2}$, the closed walk. $\square$

Lemma 3 (the step is uniform in d). *The depth step $\ell - 1 \rightarrow \ell$ adds two lattice coordinates and is a finite-order holonomic operator on the walk count for every lattice dimension h .*

Proof. By Lipshitz [2], the constant term of the N -th power of a fixed Laurent polynomial is P -recursive in N . The walk counts are such constant terms with S the fixed symmetric Laurent polynomial of $\mathbb{Z}^h$; for each fixed h the sequence in N satisfies a finite-order linear recursion with polynomial coefficients. The depth step advances $N \rightarrow N + 2$; Lipshitz guarantees finiteness for every h . This carries the uniformity in d , not the equality, which is Lemmas 1–2. $\square$

Proof of the Theorem. Fix d, ℓ . By Lemma 1, $\text{rec}(d, \ell) = \dim_{\mathbb{C}} \overline{C}_{J(2s)}(2s)$, $s = \ell - 1$. By Lemma 2 this equals the closed walk of 2ℓ steps on $\mathbb{Z}^{[d/2]}$. By Proposition 1 that walk is $\text{DS}_{2(\ell-1)}(d+1)$. Hence $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1)$, standalone at each depth. The base $\text{rec}(d, 1) = d$ anchors the ladder; Lemma 3 makes the depth step a finite uniform holonomic operator, so the identity propagates to all depths. The dictionary on which Lemma 1 rests is discharged in §6. $\square$

6 The red link — the $\mathbb{F}_p$ refinement is faithful

This section proves the faithfulness half of the dictionary (§3), discharging the one stated condition and making the block-rank law unconditional. It is pen-and-paper and characteristic-independent.

What must be shown. The faithfulness question is whether the characteristic- p tower injects extra survivors into the block closure under $\mathbb{Z} \rightarrow \mathbb{F}_p$ — equivalently, whether the integer closure module is $\mathbb{Z}$ -free, since a $\mathbb{Z}$ -free module of rank r satisfies $\mathbb{Z}^r \otimes \mathbb{F}_p = \mathbb{F}_p^r$ exactly, for *every* prime p . This is *not* the vacuous “free is free”; the object proved free is the concrete coupled closure — the image of ρ_J under the shift — by its tensor-of-cyclic-rings structure.

The mechanism. A block at depth ℓ is a tensor of ℓ within-pair closures (the partition is a perfect matching; ρ_J factorises over its pairs; the closure under shift respects the factorisation). It suffices to prove a single within-pair closure is $\mathbb{Z}$ -free.

Lemma 4 (ρ is t_A -cyclic of full dimension $m - 1$). *Let $\rho = \sum_{0 \leq b \leq a \leq m-2} x^a y^b$ be the within-pair triangle in $(\mathbb{Z}[t]/\varphi)_x \otimes (\mathbb{Z}[t]/\varphi)_y$. Then the x -shift orbit of ρ spans a submodule of dimension $m - 1$, so ρ generates the within-pair closure cyclically as a t_A -module.*

Proof. Collecting y -powers, $\rho = \sum_{b=0}^{m-2} c_b(x) y^b$ with y^0 -component $c_0(x) = 1 + x + x^2 + \dots + x^{m-2}$. In $\mathbb{Z}[t]/\varphi$ the cyclotomic relation is $x^{m-1} = -(1 + x + \dots + x^{m-2}) = -c_0$. Therefore

$$x \cdot c_0 = (x + x^2 + \dots + x^{m-2}) + x^{m-1} = (c_0 - 1) + (-c_0) = -1.$$

So $x \cdot c_0 = -1$ (verified byte-exact $m = 4 \dots 12$). The x -orbit of c_0 thus contains -1 , hence 1 , hence $x, x^2, \dots, x^{m-2}$ — it spans all of $(\mathbb{Z}[t]/\varphi)_x$. So c_0 is a cyclic vector for the x -action; being the y^0 -component of ρ , the x -orbit of ρ has full dimension $m - 1$. $\square$

This is the exact reason the within-pair closure is saturated while the image of $(t - 1)$ is *not* (the latter has Smith form $[1, \dots, 1, m]$, index m , verified $m = 4, 5, 6, 8$): the triangle ρ regenerates the whole ring ($x \cdot c_0 = -1$ pulls the unit 1 into the orbit), whereas $(t - 1)$ is a non-invertible operator whose image contracts by a p -adic factor. Saturation comes from cyclic regeneration, *not* from “the shift is a unit” — a unit-built map can have non-saturated image, so that argument would be unsound; this one is not.

Lemma 5 (the within-pair closure is $\mathbb{Z}$ -free $\cong \mathbb{Z}[t]$). *$/\varphi$ The within-pair closure, as a t -module under the shift, has characteristic polynomial exactly $\varphi = 1 + t + \dots + t^{m-1}$; hence it is isomorphic to $\mathbb{Z}[t]/\varphi$, free of rank $m - 1$.*

Proof. By Lemma 4 the module is cyclic of dimension $m - 1 = \deg \varphi$, generated by ρ . A cyclic $\mathbb{Z}[t]$ -module of dimension $\deg \varphi$ on which t reaches full dimension has characteristic polynomial φ and is $\cong \mathbb{Z}[t]/\varphi$. Verified byte-exact: characteristic polynomial of the shift on the closure $= \varphi$ for $m = 4, 5, 6$. $\mathbb{Z}[t]/\varphi$ is free as a $\mathbb{Z}$ -module (basis $1, t, \dots, t^{m-2}$). $\square$

Theorem 2 (red link — faithful refinement). *The all- A coupled block closure at depth ℓ is $\mathbb{Z}$ -free of rank $(m - 1)^\ell$. Hence $\mathbb{Z}^{(m-1)^\ell} \otimes \mathbb{F}_p = \mathbb{F}_p^{(m-1)^\ell}$ for every prime p , the $\mathbb{F}_p$ refinement injects zero torsion, and the dictionary’s faithfulness half holds.*

Proof. The depth- ℓ closure is the tensor of ℓ within-pair closures (matching factorisation of ρ_J , shift respecting it). Each within-pair closure is $\mathbb{Z}$ -free (Lemma 5). A tensor of free $\mathbb{Z}$ -modules is $\mathbb{Z}$ -free, of rank $(m - 1)^\ell$. For a free module rank determines the module up to isomorphism, so the measured rank- $(m - 1)^\ell$ match *is* the tensor (rank-match suffices precisely because both sides are free — the gap torsion would open is absent). A free $\mathbb{Z}$ -module reduces faithfully modulo every prime: $\mathbb{Z}^r \otimes \mathbb{F}_p = \mathbb{F}_p^r$, no injection. The isomorphism is over $\mathbb{Z}$ and names no prime, so it holds in every characteristic, including characteristic 2 (the $m = 8$, $\dim A = 7$ tall-tower case, where the tower is highest). $\square$

Independent verification. $x \cdot c_0 = -1$ byte-exact $m = 4..12$; characteristic polynomial of the shift on the within-pair closure $= \varphi$ for $m = 4, 5, 6$; ρ is t_A -cyclic of dimension $m - 1$ for $m = 4..10$; $\text{im}(t - 1)$ has Smith $[1, \dots, 1, m]$ for $m = 4, 5, 6, 8$; $\text{im}(\sigma)$ (the symmetriser $I + \tau$) carries 2-torsion, confirming the within-pair closure is *not* $\text{im}(\sigma)$. Four independent computational instruments measured zero injected torsion across $\dim A \in \{1, 2, 3, 7\}$ and characteristic $\in \{2, 3\}$ before the proof; the proof explains the measurements rather than resting on them.

7 Verification

Every numeric instance is recomputed from scratch.

d	derived ladder (base + the law)	target (measured)	status
2	$\text{rec}(2, 2) = 6, \text{rec}(2, 3) = 20, \text{rec}(2, 4) = 70$	6, 20, 70	byte-exact
3	$\text{rec}(3, 2) = 19, \text{rec}(3, 3) = 141, \text{rec}(3, 4) = 1107$	19, 141, 1107	byte-exact (1107 hardware)
4	$\text{rec}(4, 3) = 400, \text{rec}(4, 4) = 4900$	400, 4900	byte-exact
6	$\text{rec}(6, 4) = 44730$	$\text{DS}_6(7) = 44730$	byte-exact (dim-3 lattice)
7	$\text{rec}(7, 4) = 103279$	$\text{DS}_6(8) = 103279$	byte-exact (dim-3 lattice)

The $d = 6, 7$ rows verify the law in lattice dimension three against the published DS_6 . Beyond the published closed forms the closed walk *is* the definition.

8 Significance and scope

What is proved. The per-block rank $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d + 1)$ for every d , unconditionally, by a module-dimension identity read from the Degtyarev–Shimada source, with the one identification it rested on (the dictionary) discharged by the red link (§6).

What this does not give. The result fixes the per-block *value* — the rank ladder. It is not a statement about integral torsion. The Degtyarev–Shimada criterion compares $\dim_{\mathbb{C}}$ against $\dim_{\mathbb{F}_p}$;

this theorem computes the $\mathbb{C}$ -side ladder, and §6 proves the $\mathbb{F}_p$ refinement injects no *spurious* torsion — but neither exhibits nor excludes a *genuine* torsion counterexample in some unmeasured block. The integral torsion question — Degtyarev–Shimada Conjecture 1.2, the campaign’s “swan” — sits outside this chain and remains open. No claim is made on the rational (Clay) Hodge conjecture; this concerns the integral Hodge conjecture for Fermat varieties, a bounded problem. The torsion counterexample has not been found; every cell measured is primitive. What the red link gives the hunt is a rigorous map: torsion lives in one block (localization), the per-block value is fixed and the refinement is faithful, so a counterexample can only hide in a recursive-dimension ≥ 4 block deviating from the Degtyarev–Shimada value; recdim 0, 2 are torsion-free by [1, Cor. 1.7].

References

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